

What Is Covariance?

Covariance is a statistical term that refers to a systematic relationship between two random variables in which a change in the other reflects a change in one variable.

The covariance value can range from $-\infty$ to $+\infty$, with a negative value indicating a negative relationship and a positive value indicating a positive relationship.

The greater this number, the more reliant the relationship. Positive covariance denotes a direct relationship and is represented by a positive number.

A negative number, on the other hand, denotes negative covariance, which indicates an inverse relationship between the two variables. Covariance is great for defining the type of relationship,

but it's terrible for interpreting the magnitude.

Let $\Sigma(X)$ and $\Sigma(Y)$ be the expected values of the variables, the covariance formula can be represented as:

$$\text{Covariance}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Where,

- x_i = data value of x
- y_i = data value of y
- $\bar{x}$ = mean of x
- $\bar{y}$ = mean of y
- N = number of data values.

Types of Covariance

Covariance can be either positive or negative. It is classified into two sorts based on this:

Positive Covariance

A positive covariance between two variables indicates that they are heading in the same direction. The variables, in this case, behave similarly. That is, if the values of one variable (more or smaller) correspond to the values of another variable, they are said to be in positive covariance.

Negative Covariance

When two variables have a negative covariance, the variables shift in the opposite direction. It is the inverse of positive covariance, in which higher values of one variable correlate to lower values of another and vice versa.

Application Of Covariance

The following are the most common applications of Covariance:

- Simulating systems with multiple correlated variables is done using Cholesky decomposition. A covariance matrix helps determine the Cholesky decomposition because it is positive semi-definite. The matrix is decomposed by the product of the lower matrix and its transpose.
- To reduce the dimensions of large data sets, principal component analysis is used. To perform principal component analysis, an eigen decomposition is applied to the covariance matrix

What Is Correlation?

In statistics, correlation is a measure that determines the degree to which two or more random variables move in sequence. When an equivalent movement of another variable reciprocates the movement of one variable in some way or another during the study of two variables, the variables are said to be correlated. The formula for correlation is:

$$\rho_{xy} = \text{Correlation}(x, y) = \frac{\text{cov}(x, y)}{\sqrt{\text{var}(x)}\sqrt{\text{var}(y)}}.$$

where,

$\text{var}(X)$ = standard deviation of X

$\text{var}(Y)$ = standard deviation of Y

Positive correlation occurs when two variables move in the same direction. When variables move in the opposite direction, they are said to be negatively correlated.

Correlation is of three types:

1. Simple Correlation: In simple correlation, a single number expresses the degree to which two variables are related.
2. Partial Correlation: When one variable's effects are removed, the correlation between two variables is revealed in partial correlation.
3. Multiple correlation: A statistical technique that uses two or more variables to predict the value of one variable.

Methods of Calculating Correlation

There are a number of methods to calculate correlation coefficient. Here are some of the most common ones:

Coefficient of correlation

This is the most common method of determining the correlation coefficient of two variables. It is obtained by dividing the covariance of two variables with the product of their standard deviations.

$$r = \frac{\text{Cov}(x,y)}{\sqrt{\text{Var}(x) \text{Var}(y)}}$$

For other formulas of correlation and properties, consult recommended text book